

SEMIH KAVAK

Software Architect | Tech Lead | Staff Software Engineer

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PROFESSIONAL SUMMARY

Software engineer with 14+ years of experience designing and developing enterprise-grade applications, backend platforms, and distributed systems, with deep expertise in Java and the Spring ecosystem. Strong background in microservices architecture, system integration, messaging-driven systems, CI/CD, and container-based deployments across complex enterprise and product environments.

Experienced in combining hands-on engineering with technical leadership, architecture contribution, mentoring, and cross-team collaboration. Proven ability to work across backend, full-stack, DevOps-related delivery processes, and platform modernization initiatives. Currently targeting Software Architect, Tech Lead, and Staff Software Engineer roles where architectural ownership and long-term technical impact are valued.

CORE COMPETENCIES

- Software Architecture
- Technical Leadership
- Java & Spring Ecosystem
- Microservices Architecture
- Distributed Systems
- RESTful API Design
- Event-Driven Systems
- System Integration
- DevOps Collaboration
- CI/CD and Release Engineering
- Docker / Kubernetes / OpenShift
- Messaging Systems
- Linux-Based Environments
- Performance and Reliability Improvements
- Code Review and Mentoring
- Agile / Scrum Delivery

TECHNICAL EXPERTISE

Architecture & Platforms:

Software Architecture, Microservices Architecture, Distributed Systems, Event-Driven Systems, Enterprise Application Development, System Integration, RESTful APIs, SOAP Services, Design Patterns, OOP, UML

Backend Engineering:

Java 8/17+, Spring Framework, Spring Boot, Spring Cloud, Hibernate, JPA, iBatis, Java Concurrency, Multi-Threading, Embedded Jetty

Frontend & Full Stack:

JavaScript, React, Angular, Node.js, JSF, PrimeFaces, IceFaces, GWT, MyFaces

Messaging & Integration:

Kafka, RabbitMQ, ActiveMQ, Mule ESB, JAX-WS

Databases & Persistence:

PostgreSQL, MySQL, MSSQL, Oracle DB, MongoDB, PL/SQL, Stored Procedures

DevOps & Delivery:

Docker, Kubernetes, OpenShift, Jenkins, Helm, CI/CD, Linux, Nginx, Git, GitHub, SVN

Monitoring & Reporting:

Elasticsearch, Grafana, JasperReports, Highcharts, Hazelcast

Specialized Areas:

Industrial IoT, Embedded Software, Embedded Linux, Augmented Reality, Android Development, WebRTC

Tools & Practices:

IntelliJ IDEA, Eclipse, NetBeans, Maven, Gradle, Jira, Agile, Scrum, Code Review, Mentoring, Technical Leadership, Production Support, SDLC, DevOps Collaboration

PROFESSIONAL EXPERIENCE

Tech Lead & Expert Software Engineer

Turkish Technology | Sep 2024 – Present

- Lead a team of 4–5 engineers in designing, developing, and maintaining scalable microservice-based backend systems.
- Own technical decision-making, code quality practices, architecture alignment, and mentoring of less experienced engineers.
- Designed and implemented backend services using Java 17+, Spring Boot, PostgreSQL, Kafka, RabbitMQ, Redis, and Elasticsearch.
- Improved deployment reliability and delivery consistency through CI/CD pipelines built with Jenkins and OpenShift.
- Contributed actively to containerization, Helm-based deployment flows, and OpenShift environment maintenance as part of engineering delivery.
- Collaborated with product owners and stakeholders to convert business needs into maintainable and production-ready technical solutions.

Technologies: Java 17+, Spring Boot, Microservices, PostgreSQL, Kafka, RabbitMQ, Redis, Elasticsearch, Docker, Jenkins, OpenShift, Angular, React, Git, CI/CD, Agile/Scrum

Senior Full Stack Software Engineer

HAVELSAN | Aug 2022 – Aug 2024

- Worked on classified enterprise software projects focused on modern web systems and microservice-based architectures.
- Contributed to the design and implementation of backend and full-stack components using Java-based technologies and modern frontend stacks.
- Participated in DevOps and engineering enablement tasks including Git integration, Jenkins pipelines, and CI/CD process improvements.
- Supported delivery of scalable Linux-based systems in technically demanding enterprise environments.

Technologies: Java 8+, Spring, Spring Boot, Hibernate, REST APIs, MySQL, PostgreSQL, Docker, Kubernetes, React, Node.js, Git, Jenkins, Linux, CI/CD, Microservice Architecture

Senior Backend Software Engineer

eBay Turkey / GittiGidiyor | Mar 2022 – Aug 2022

- Contributed to e-commerce backend services related to user operations, cart management, and pre-payment workflows.
- Worked on microservice-based systems using Java, Spring, Hibernate, Hazelcast, and containerized deployment environments.
- Supported engineering delivery processes through CI/CD work, Git-based development workflows, and Jenkins integration.
- Contributed to the stability and maintainability of business-critical services in a fast-paced product environment.

Technologies: Java 8+, Spring, Spring Boot, Hibernate, MySQL, Hazelcast, Docker, Kubernetes, Git, Jenkins, Linux, REST APIs, Microservice Architecture, CI/CD

Senior Full Stack Software Engineer

ASELSAN | Jan 2020 – Mar 2022

Rail Systems Projects

- Worked on classified rail systems projects involving both web applications and embedded software components.
- Contributed to development and engineering delivery processes across C++, Java, C#, and Node.js-based components.
- Supported CI/CD-related work including Git workflows and Jenkins integrations.

Technologies: C++, Embedded Linux, Java 8+, Node.js, Git, Jenkins

Industrial IoT Projects

- Contributed to Industrial IoT and microservice-based platforms running in containerized environments.
- Worked on Java-based services deployed first on vanilla Kubernetes and later on OpenShift infrastructure.
- Supported Git and Jenkins integration efforts as well as CI/CD-related engineering improvements.
- Participated in backend platform development together with modern frontend integration for enterprise web systems.

Technologies: Java 8+, Spring Boot, Microservices, PostgreSQL, Elasticsearch, RabbitMQ, Docker, Kubernetes, OpenShift, React, Angular, Keycloak, JasperReports, Grafana, Git, Jenkins, Linux, CI/CD

Java Software Engineer

Qualist (Pratikkod Yazılım Geliştirme A.Ş.) | Apr 2013 – Jan 2020

Worked across multiple industrial software products in monitoring, analytics, workflow management, remote support, and operational intelligence.

GreenChimney

- Contributed to a real-time emissions monitoring platform used to collect analyzer data, calculate regulatory averages, generate reports, manage alarms, archive operational data, and communicate with public authority systems.
- Worked on backend-heavy Java development in a system requiring concurrency, reliability, reporting, and large-scale industrial data handling.

Technologies: Java 8+, Spring Boot, Microservices, REST APIs, MySQL, MongoDB, RabbitMQ, Hazelcast, Docker, Jenkins, GWT, JasperReports, Highcharts, Tomcat, Linux

Qualist Bridge

- Contributed to industrial analytics software used for quality improvement, preventive maintenance, production optimization, and utility efficiency.
- Worked on scalable business applications serving manufacturing and operational analytics use cases.

Technologies: Java 8+, Design Patterns, MySQL, MSSQL, GWT, Hazelcast, JasperReports, Highcharts, Git, Jenkins, Tomcat, Linux, Windows

Recipe Management

- Worked on a customer-specific workflow and recipe management solution built for industrial environments.
- Contributed to systems collecting data from PLCs and OPC servers and enabling customers to manage operational workflows.

Technologies: Java 8+, Monolith Architecture, Design Patterns, MySQL, MSSQL, Java Concurrency, JasperReports, Highcharts, Git, Jenkins, Tomcat, Linux, Windows

Qualist Remote Assist

- Contributed to an augmented reality-based remote support product that enabled field workers to access maintenance documentation and receive live assistance using mobile devices or smart glasses.
- Worked in an innovation-focused product area combining enterprise software engineering with AR-enabled workflows.

Technologies: Java 8+, Augmented Reality, Android, WebRTC, MySQL, GWT, S3, Git, Jenkins, CI/CD

Java Software Engineer

Sampaş A.Ş. | Feb 2012 – Apr 2013

- Developed Java web services and enterprise backend components for public-sector and utility-related projects.
- Worked on report modernization and development using Oracle BI Publisher.
- Contributed to e-municipality systems, smart meter integrations, and notification/messaging services under international project scope.
- Built SOAP and REST-based services and worked extensively with Oracle database integrations, stored procedures, and Mule ESB.

Technologies: Java, Spring, iBatis, Hibernate, JSF, IceFaces, PrimeFaces, JAX-WS, SOAP, REST APIs, Oracle DB, Stored Procedures, Mule ESB, Oracle BI Publisher

INTERNSHIPS

Computer Engineer Intern

Airties Wireless Network | Aug 2011 – Sep 2011

- Developed a C++ desktop application and supported device testing through Telnet-based connectivity.

Computer Engineer Intern

Koç Finans A.Ş. | Jul 2011 – Aug 2011

- Contributed to web page development using MyFaces and IceFaces.

Computer Engineer Intern

Arttek A.Ş. | Jul 2010 – Aug 2010

- Worked on website development using PHP, MySQL, CSS, and Ajax.

EDUCATION

Istanbul Technical University (ITU)

M.Sc. in Computer Engineering (English) | 2012 – 2016

Left at thesis stage

Istanbul Technical University (ITU)

B.Sc. in Computer Engineering (English) | 2007 – 2012

GPA: 3.28 / 4.00

CERTIFICATIONS

- ISO Education Certificate — TÜV Rheinland, Oct 2017
- Project Management in Engineering — Turkcell, Jun 2011

TRAINING

- Communication Training — Gökhan Karahan, 32 hours, 2016
- Docker Training — Kodcu.com, 40 hours, 2016